



Hamza Bouktitiya

Artificial Intelligence Engineer • PhD Candidate

📍 Kenitra, Morocco • 📞 +212 688 679 653 • bouktitiya.hamza.post@gmail.com
[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

AI engineer and PhD candidate building production LLM systems (RAG, agents, APIs) with FastAPI, LangChain/LangGraph, and Docker. Experience spanning NLP, computer vision, and full-stack deployment.

Education

- PhD Candidate in Artificial Intelligence** 2024 – Present
Ibn Tofail University, Kenitra, Morocco
Thesis: *Textual description and automatic classification of video sequences by deep learning.* Supervisor: Prof. Housni Khalid.
- Master in Artificial Intelligence and Virtual Reality** 2022 – 2024
Ibn Tofail University, Kenitra, Morocco
- Bachelor in Mathematical and Computer Science** 2018 – 2022
Ibn Tofail University, Kenitra, Morocco

Professional Experience

- Junior AI Engineer** April 2025 – October 2025
OBYSTECH SOLUTIONS, Rabat
Designed and shipped a multi-department AI agent platform (HR, Finance, Management, IT) as a solo project.
 - Built a multi-chat Next.js front end with collection selector, per-chat RAG toggle, team workspaces, and cursor-based pagination.
 - Implemented a RAG pipeline (PDF/docs ingestion, Hugging Face embeddings, hybrid/vector search with MMR) and LangGraph workflows with PostgreSQL-backed memory; MCP for external tools.
 - Deployed a multi-service Docker Compose stack on VPS with reverse proxy, SSL, and zero-downtime restarts.
 - Stack:** Next.js, FastAPI, LangChain, LangGraph, MCP, Ollama, PostgreSQL, ChromaDB, Docker, JWT/OAuth2.
- AI Intern** February 2024 – August 2024
Verve Technologie, Casablanca
 - Built an LSTM forecasting system for energy consumption prediction.
 - Exposed Stable Diffusion 1.5 image-generation endpoints and a RAG pipeline over Wikipedia/PDF sources.
 - Added JWT authentication and user management on FastAPI.
 - Stack:** LSTM, LangChain, FastAPI, Stable Diffusion, JWT.
- Web Development Intern** March 2022 – June 2022
CS Department, Ibn Tofail University, Kenitra
Developed a gym management web app for clients and administrators (PHP, JavaScript, Bootstrap, MySQL).

Technical Skills

AI / ML	PyTorch, TensorFlow, LangChain, LangGraph, CrewAI, RAG, LLMs, NLP, Computer Vision, NeRF
Back-end	FastAPI, Django, Python, REST APIs, PostgreSQL, MySQL, JWT/OAuth2
Front-end	Next.js, React, TypeScript, Tailwind CSS, HTML/CSS, Streamlit
MLOps / Ops	Docker, Docker Compose, Nginx, VPS deployment, Git/GitHub
Data	Pandas, NumPy, Power BI, TF-IDF, classical ML (SVM, KNN)

Selected Projects

- Intelligent Assessment System** (2026) — LLM/RAG platform for exam generation, answer evaluation, and personalized feedback (BLEU, ROUGE, BERTScore); FastAPI, Streamlit, LangChain, Docker.
- Computer Vision Playground** (2025) — Dockerized Streamlit app for MTCNN face detection, LBPH recognition, YOLOv8 object detection, and MediaPipe pose estimation.
- Darija Q&A System** (2023) — Bilingual Moroccan Darija/English QA with Transformer QA, Seq2Seq, and LSTM translation models.
- License Plate Recognition** (2023) — HOG features with KNN/SVM character classifiers; **98.9%** test accuracy.

Languages & Activities

Languages: Arabic (Native) • English (C1) • French (B2)
Activities: DyslexAI Hackathon — Top 3 (Al Akhawayn University) • ThinkAI Hackathon (1337 Coding School) • Co-founder, Pragnomos Club • ML talk at Agora event